

Question Number	Answer	Additional guidance	Mark
6(a)	An answer that includes the following points: - the extract {was effective / had anti-microbial properties} against all three types of bacteria (1) - as it {prevented growth of / killed} bacteria (around the disc soaked in extract) (1) - it was most effective against bacteria A / least effective against bacteria B (at both concentrations) (1)	accept positive correlation accept higher concentration of extract has more antimicrobial properties accept as the (extract) concentration increases, the {diameter / zone of inhibition} increases accept it inhibited (the growth of) A the most / B had most resistance	(3)

Question Number	Answer	Additional guidance	Mark
6(b)(i)	An answer that includes four of the following points: • as concentration increased the duration of headaches decreased / the lower the concentration of the drug the higher the duration of headaches (1) • no data for 50 and 100mg kg⁻¹ for drug B / no data below 200 mg kg⁻¹ for drug B / drug B tested over smaller range of concentrations (1) • the duration of headaches was shorter with drug A (than with drug B) / drug A is more effective (at reducing duration of headaches) (1) • can't say which drug is more effective due to overlap of {error/ SD} bars (1) • relevant comment about methodology (1)	accept negative correlation accept drug B was only tested for {the range 200-400 mg kg⁻¹ /above 200 mg kg⁻¹ / 2 concentrations} ignore drug B doesn't work below 200 mg kg⁻¹ ignore there are no headaches below 200 mg kg⁻¹ with drug B ACCEPT converse for drug B accept there is {no significant difference / low validity / low reliability} as {error / SD} bars overlap ignore size of error bars e.g. lack of information about controlled variables / sample size / age	(4)

Question Number	Answer	Additional guidance	Mark
6(b)(ii)	An answer that includes three of the following points: Similarities (max 2) • both test on patients with headaches (1) • both can include {placebos / double-blind testing} (as well as drug A) (1) • results from both phases would be {analysed with appropriate statistical test / tested for significant difference to placebo} (1) Differences • (drug A is) tested on {small group of / 100 to 500} people in phase II whereas III tested on {large groups of / 1000-3000} people (1) • appropriate {concentration/ dosage} (of drug A) would be identified in phase II (but not in phase III) (1)	Can only piece together from adjacent sentences accept both test on affected patients / both test on patients with the {disease / symptom} ignore patients unqualified accept both look for side effects accept III tested on larger group (than II) / converse accept {effective dose found / different concentrations used/ check dosage / determine dosage} in phase II	(3)